Modelling of Dust Settling in Protoplanetary Discs via Advection–Diffusion Numerical Methods

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***Abstract*—We investigate the vertical settling of dust in a protoplanetary disc using a one-dimensional advection–diffusion model to determine whether the formation of planetesimals is possible. The dust evolution equations are solved numerically using a finite-difference scheme. By comparing our simulations with observational constraints, we estimate the turbulence parameter to be $\alpha \sim 10^{-4}$. We also estimate that dust particle sizes of approximately 50–$200\,\mu\mathrm{m}$ are required for planetesimal formation within our advection–diffusion model. Finally, we show that our results are largely unaffected by numerical diffusion.**

I. Introduction

UNDERSTANDING how planets form around young stars remains one of the central problems in modern astrophysics. Planets are believed to form in flattened, rotating structures known as protoplanetary discs, composed primarily of gas with a small admixture of solid dust particles. These dust grains act as the building blocks of planetesimals, kilometre-scale bodies that are thought to be the precursors of planets. A key requirement for planetesimal formation is that the local dust-to-gas density ratio exceeds unity, enabling collective instabilities such as the streaming instability to operate.

While the gas in a protoplanetary disc is supported vertically by pressure and remains in hydrostatic equilibrium, dust particles experience vertical gravity and tend to settle toward the disc midplane. In cylindrical coordinates (R, z), assuming axisymmetry and symmetry about the midplane $z = 0$, the gas density profile is given by

$$\rho_g(z) = \rho_g(R, 0) \exp\left(-\frac{z^2}{2H(R)^2}\right), \qquad (1)$$

where $H(R) = c_s/\Omega$ is the disc scale height, c_s is the sound speed, and Ω is the Keplerian angular frequency.

The vertical evolution of the dust density ρ_d can be modelled as a flux-conservative advection process,

$$\frac{\partial \rho_d}{\partial t} + \frac{\partial}{\partial z}(u_z \rho_d) = 0, \qquad (2)$$

where the dust settles toward the midplane with a terminal velocity

$$u_z = -\Omega^2 z\, t_s, \qquad (3)$$

and the stopping time t_s depends on the dust grain size and local gas properties. In this framework, dust settling can lead to a strong enhancement of the midplane dust density over time.

However, observations indicate that micron-sized dust particles remain vertically extended in the outer regions of discs even after ~ 1 Myr, suggesting that turbulent mixing counteracts settling. Turbulence is commonly modelled as an effective diffusion with coefficient

$$D = \alpha \frac{c_s^2}{\Omega}, \qquad (4)$$

where α is a dimensionless parameter characterising the turbulence strength. Including this effect, the dust evolution obeys an advection–diffusion equation,

$$\frac{\partial \rho_d}{\partial t} + \frac{\partial}{\partial z}(u_z \rho_d) = \frac{\partial}{\partial z}\left[D \rho_g \frac{\partial}{\partial z}\left(\frac{\rho_d}{\rho_g}\right)\right]. \qquad (5)$$

In this project, we solve these equations numerically using a one-dimensional finite-difference scheme. We first verify the numerical method and assess the impact of numerical diffusion, before modelling dust settling with and without turbulence. By comparing simulations with observational constraints at $R = 100\,\mathrm{AU}$, we estimate the turbulence parameter α and examine its effect on the ability of dust to form planetesimals. Finally, we explore how dust settling depends on particle size and radial distance in regions where giant planets are observed to form.

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II. Numerical Method

We solve the dust evolution equations in one spatial dimension using finite-difference methods. All quantities are expressed in dimensionless form, with $\hat{z} = z/H(R)$ and $\hat{t} = \Omega t$. The dust density $\rho_d(\hat{z}, \hat{t})$ is evolved on a uniform grid using explicit time stepping, subject to stability constraints.

A. Advection

In the absence of turbulence, the dust evolves according to the dimensionless advection equation

$$\frac{\partial \rho_d}{\partial \hat{t}} + \frac{\partial}{\partial \hat{z}} \left(u(\hat{z})\, \rho_d \right) = 0, \tag{6}$$

where $u(\hat{z}) \leq 0$ is the settling velocity towards the midplane. We discretise this equation using a first-order upwind scheme on a uniform grid with spacing $\Delta\hat{z}$ and explicit Forward Euler time stepping. The advective flux is computed as

$$F_i = u_i \rho_{d,i}, \tag{7}$$

and each interior cell is updated via

$$\rho_{d,i}^{n+1} = \rho_{d,i}^{n} - \frac{\Delta\hat{t}}{\Delta\hat{z}} \left(F_{i+1}^{n} - F_i^{n} \right). \tag{8}$$

Because the velocity is negative, this corresponds to a first-order upwind method and is first-order accurate in space and time.

A reflective boundary at the midplane ($\hat{z} = 0$) is imposed using a ghost cell with mirrored density and sign-reversed velocity, ensuring zero net flux across the midplane. At the upper boundary, the ghost-cell density and velocity are set to zero, allowing dust to leave the domain without inflow. Ghost cells are applied at every timestep before computing the flux differences.

B. Diffusion

Including turbulent diffusion gives the dimensionless PDE

$$\frac{\partial \rho_d}{\partial \hat{t}} + \frac{\partial}{\partial \hat{z}}(u\rho_d) = \alpha \, \frac{\partial}{\partial \hat{z}} \left[\rho_g(\hat{z}) \, \frac{\partial}{\partial \hat{z}} \left(\frac{\rho_d}{\rho_g} \right) \right], \tag{9}$$

where $\rho_g(\hat{z})$ is the fixed gas density profile and α is the diffusion parameter. We evaluate the diffusion term in flux form, defining

$$F^{(D)} = -\alpha \, \rho_g(\hat{z}) \frac{\partial}{\partial \hat{z}} \left(\frac{\rho_d}{\rho_g} \right), \tag{10}$$

so that the diffusion update is the negative divergence of this flux.

On the grid, we compute $f_i = \rho_{d,i}/\rho_{g,i}$ at cell centres, approximate ρ_g at faces by arithmetic averaging, and compute the gradient using centred differences. The diffusive flux divergence is then used to update the interior cells. Zero diffusive flux is imposed at both boundaries by setting the ghost-face fluxes to zero. At the midplane, one additional update involving the lower ghost face ensures the reflective symmetry is satisfied.

C. Grid, timestep, and stability

The domain extends from $\hat{z} = 0$ to $\hat{z}_{\max} = 3.0$ and is discretised using N uniformly spaced points, with $\Delta\hat{z} = \hat{z}_{\max}/(N-1)$. Values of $N = 200$–600 provide a balance between accuracy and computational cost.

Time integration uses explicit Forward Euler stepping with timestep $\Delta\hat{t}$. Stability requires both the advective CFL condition

$$|u| \, \frac{\Delta\hat{t}}{\Delta\hat{z}} < 1, \tag{11}$$

and the diffusive constraint

$$\Delta\hat{t} \leq \frac{(\Delta\hat{z})^2}{2\alpha}. \tag{12}$$

A single timestep satisfying both restrictions is chosen for each simulation.

D. Boundary conditions and numerical diffusion

Ghost cells are used to impose both the midplane and upper boundary conditions. At the midplane ($\hat{z} = 0$), the dust density is mirrored and the velocity sign-reversed, producing a reflective, zero-flux boundary. For diffusion, the diffusive flux is also set to zero here to enforce symmetry. At the upper boundary ($\hat{z} = \hat{z}_{\max} = 3.0$), the ghost-cell density and velocity are set to zero, providing an outflow boundary for advection and a zero-flux condition for diffusion.

Because the upwind advection scheme is first-order accurate, it introduces artificial numerical diffusion that smooths sharp gradients when $\alpha = 0$. This effect decreases with grid refinement and is negligible compared with the physical diffusion when $\alpha > 0$.

III. Results

A. *Advection-only evolution ($\alpha = 0$)*

In the absence of turbulent diffusion, dust settles purely under the vertical advection term. For $R_{\rm AU} = 1$, a grain size of $1.0\,\mu$m, $N = 400$, and $\hat{z}_{\rm max} = 3.0$, the dust profile progressively steepens as particles drift toward the midplane. The midplane density increases monotonically, and the dust layer becomes progressively thinner, reflecting the continual concentration of material into a narrow region around $\hat{z} = 0$ (Fig. 1).

Because no physical diffusion is present, the ideal solution would develop an increasingly sharp peak at the midplane. In practice, the first-order upwind advection scheme introduces numerical diffusion that smooths the peak and slightly broadens the dust layer at late times. This effect becomes noticeable after many timesteps and is reduced when the grid resolution is increased, consistent with the expected first-order truncation behaviour.

The dust-to-gas ratio near the midplane initially remains well below unity; after $1\,$Myr, it reaches only $\epsilon \approx 0.02$. However, continued settling over long timescales leads to much stronger concentration. Extending the simulation to $10\,$Myr shows that $\epsilon = 1$ is exceeded after approximately $7.4\,$Myr, illustrating that in the absence of any turbulent mixing, the dust layer can become extremely dense (Fig. 2).

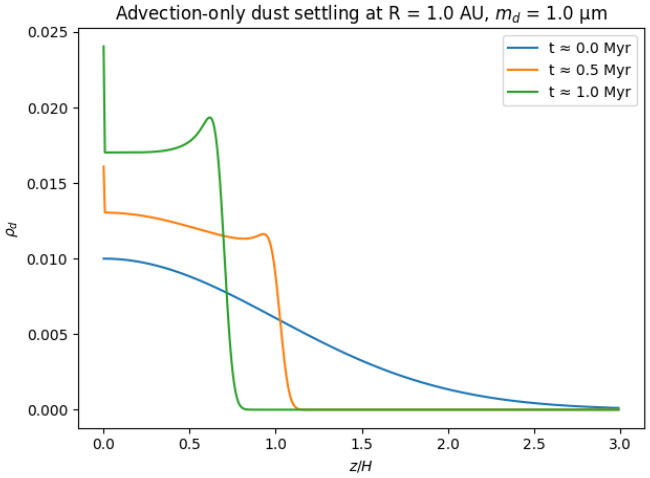


Fig. 1. Advection-only dust settling at $R = 1.0\,$AU with grain size $m_d = 1.0\,\mu$m. The dust layer steepens and concentrates toward the midplane over time. Numerical diffusion from the first-order upwind scheme smooths the peak at late times.

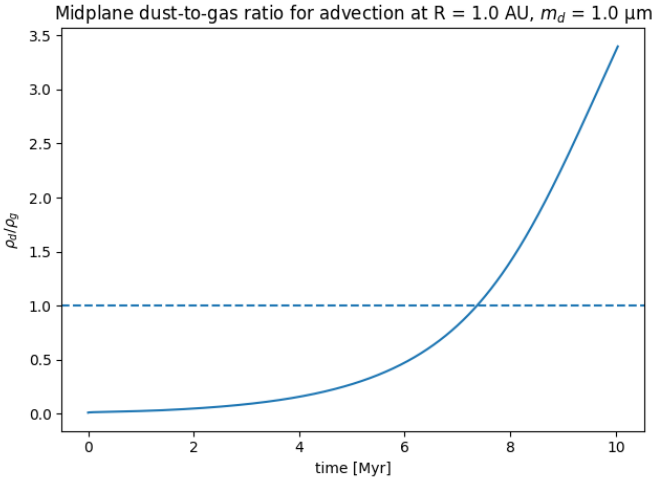


Fig. 2. Midplane dust-to-gas ratio for advection-only evolution at $R = 1.0\,$AU and $m_d = 1.0\,\mu$m. The ratio increases monotonically as dust settles toward the midplane, eventually exceeding unity after $\sim 7.4\,$Myr.

B. *Estimating the diffusion parameter α at 100 AU*

To estimate the turbulence level in the disc, we use the observational result from light-scattering imaging [1] that micron-sized dust at $R = 100\,$AU is found at a height of about $z = 0.1R$. In our model, this corresponds to roughly one gas scale height ($z/H \approx 1$). We therefore look for the value of α that produces a dust layer with a similar thickness once settling and diffusion reach a steady state on Myr timescales.

When α is very small (10^{-5}), the dust layer becomes too thin and collapses more rapidly toward the midplane. When α is large (10^{-3}), the dust spreads too widely and extends over too large of a domain.

The case $\alpha = 10^{-4}$ lies between these limits and produces a dust layer with a thickness matching the observed $z/H \approx 1$. From this we estimate

$$\alpha \sim 10^{-4}.$$

This turbulent simulation also behaves very differently from the advection-only model. With no diffusion, dust keeps settling and the midplane dust-to-gas ratio rises steadily and indefinitely. With

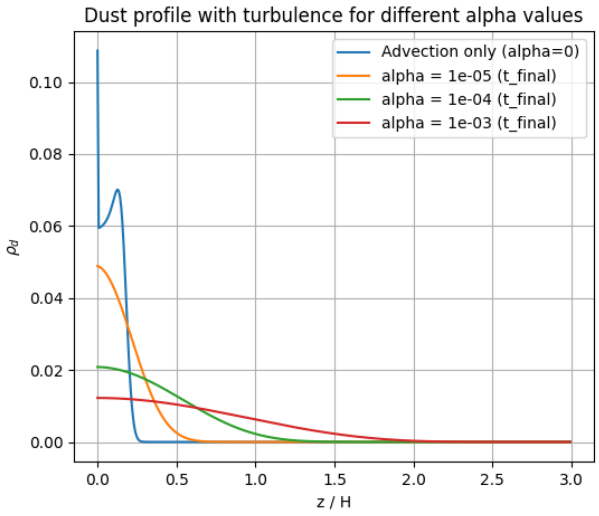


Fig. 3. Final dust density profiles at $R = 100\,\mathrm{AU}$ for $\alpha = 10^{-5}$, 10^{-4}, 10^{-3}, compared with the advection-only case. Weak turbulence (10^{-5}) produces a very thin midplane layer, while strong turbulence (10^{-3}) mixes the dust over several scale heights. $\alpha = 10^{-4}$ gives a thickness consistent with the observed $z/H \approx 1$.

$\alpha = 10^{-4}$, the ratio stays much lower and reaches only $\epsilon \approx 0.02$ in our simulations, far below the plantesimal formation threshold $\epsilon = 1$. This shows that even small turbulence prevents the formation of a very thin, dense midplane layer.

Studies typically estimate turbulence levels in the range $\alpha \sim 10^{-3} - 10^{-2}$ [1], which are higher than our estimate. This is expected given the simplicity of our model (1D, fixed gas background, single grain size). Within these limitations, $\alpha \sim 10^{-4}$ provides the closest match to the observed dust height at 100 AU.

The comparison between the advection-only case and advection-diffusion case is insightful. For a long enough time period, the advection-diffusion case reaches a steady state and a limiting dust-to-gas ratio ϵ. However with advection-only a steady state is never reached and the dust-to-gas ratio keeps increasing indefinitely, as shown in Figure 2. This means that the presence of diffusion heavily influences whether or not the system can form planetesimals.

C. Dependence on particle size and radius

So far we have focused on $1\,\mu\mathrm{m}$ dust particles at $R = 100\,\mathrm{AU}$. We now explore where planets are expected to form by varying the particle size at different radii corresponding to the orbits of the giant planets in our Solar System, namely $R = 5$, 10, 20, and $30\,\mathrm{AU}$. We repeat the simulations for a range of grain sizes with $\alpha = 10^{-4}$ and examine whether the condition for planetesimal formation, $\epsilon > 1$, is satisfied.

For small grains ($a \sim 1\,\mu\mathrm{m}$), the dust remains strongly coupled to the gas at all radii, leading to weak settling, a relatively thick dust layer, and midplane dust-to-gas ratios well below unity. Increasing the particle size to $a \sim 50$–$200\,\mu\mathrm{m}$ enhances settling and produces thinner dust layers, with the effect becoming more pronounced at smaller radii. The figure below shows, for each radius considered, the minimum particle size required for the midplane dust-to-gas ratio to exceed unity ($\epsilon > 1$).

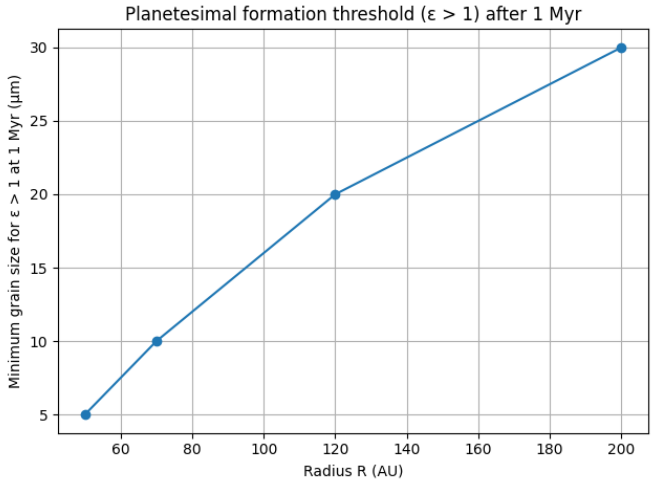


Fig. 4. Minimum dust particle size required for the midplane dust-to-gas ratio to exceed unity ($\epsilon > 1$) as a function of radius, for $\alpha = 10^{-4}$. Larger particles are required at larger radii, reflecting weaker settling and stronger coupling to the gas in the outer disc.

D. Numerical diffusion

Before analysing the physical problem, we investigated the numerical diffusion introduced by the first–order upwind advection scheme. This was tested using two initial condition profiles: a Gaussian profile and a step function. In both cases, only advection was included, so any smoothing of the profile is entirely numerical in origin.

In both tests, numerical diffusion was clearly present. The profiles exhibited smoothing, broadening, and a reduction in amplitude over time. The effect was significantly stronger for the step–function initial condition (Fig. 5), which can be

attributed to the sharp discontinuity in the profile, making it more susceptible to numerical diffusion.

To assess the impact of numerical diffusion on the full advection–diffusion simulations, we repeated the calculations while doubling the number of grid points. Figure 6 shows that the change in the midplane dust–to–gas ratio, $\mathrm{d}\epsilon/\mathrm{d}N$, decreases and begins to level off as the resolution increases, suggesting that the solution is approaching convergence. Importantly, increasing the grid resolution does not change the qualitative behaviour of the solution; the overall dust distribution and trends remain the same, indicating that numerical diffusion does not affect the main physical conclusions. A natural extension of this work would be a more systematic convergence study to further quantify the influence of numerical diffusion.

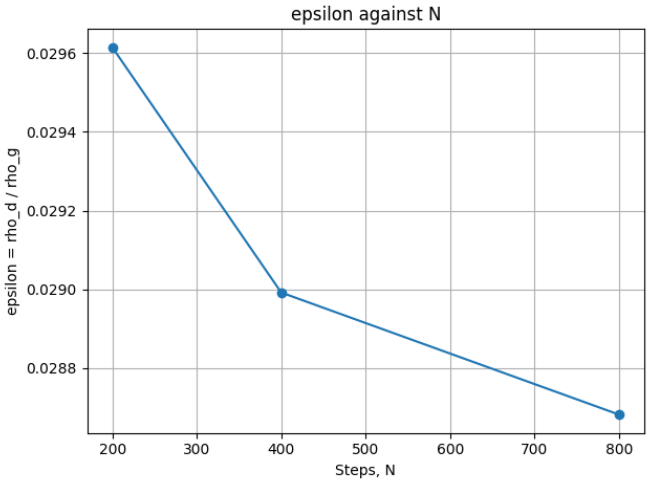


Fig. 6. Resolution test for the advection–diffusion simulations. Doubling the number of grid points reduces the change in the midplane dust–to–gas ratio, and $\mathrm{d}\epsilon/\mathrm{d}N$ decreases with increasing N, indicating the solution is approaching convergence.

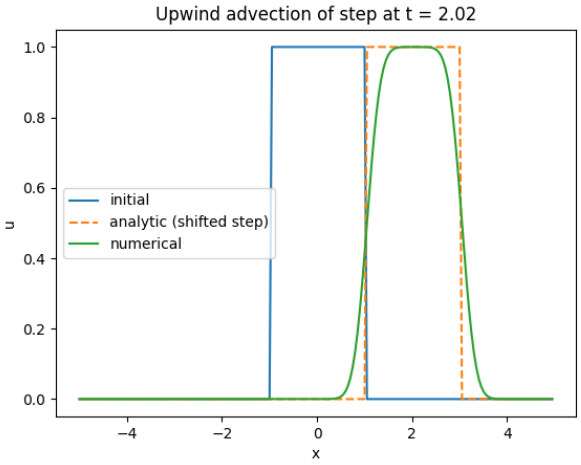


Fig. 5. Advection of a step–function initial condition using the first–order upwind scheme. The discontinuity becomes rounded and spreads over multiple grid cells over time, demonstrating numerical diffusion.

IV. Discussion

The results show that dust settling and planetesimal formation are strongly regulated by turbulent diffusion. In the absence of diffusion, high dust-to-gas ratios are reached rapidly; however, this behaviour is unlikely in realistic protoplanetary discs.

By matching our simulated dust profiles to observations at $R = 100\,\mathrm{AU}$ [2], we estimate a turbulence parameter of $\alpha \sim 10^{-4}$. At this value, the dust layer width in our model is consistent with observations. This level of turbulence suppresses strong midplane concentration and implies that dust particle sizes of approximately 50–200 μm are required for planetesimal formation, compared to the initial assumption of micron-sized grains.

Overall, turbulence plays a key role in regulating dust concentration, and additional mechanisms beyond vertical settling are required for planetesimal formation, as dust sizes of 50–200 μm may be difficult to achieve without grain growth or other concentration processes.

V. Conclusion

We have modelled the vertical settling of dust in a protoplanetary disc using a numerical advection–diffusion approach. Numerical diffusion was shown not to affect the qualitative behaviour of the solutions. By matching simulations to observational constraints at 100 AU, we estimated a turbulence parameter of $\alpha \sim 10^{-4}$.

We further showed that planetesimal formation is favoured at radii corresponding to the giant planets in the Solar System when dust particle sizes reach approximately 50–200 μm. Smaller grains are too strongly coupled to the gas for efficient settling. These results show that turbulent diffusion suppresses midplane dust concentration, and that additional physical processes are required to enable planetesimal formation in realistic protoplanetary discs.

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